PLANT DISEASE DETECTION SYSTEM

SUMMARY

Plant disease detection system is system where we can find the disease of the plant by using its leaf image using CNN. By reading all these paper most of the researchers are using Artificial Intelligence, particularly Convolutional Neural Networks (CNNs), to automatically detect plant diseases from leaf images. Manual inspection is slow and needs more knowledge about the plant and all about its details. These systems use methods like transfer learning and data augmentation to improve accuracy and manage different lighting and angles. Transfer learning is an AI technique where a model trained on one task is reused as the starting point for a related, new task and data augmentation is a technique used to artificially create more training data for AI models by making slight, modified copies of the data you already have. The studies involve many crops, including rice, potato, tomato, and tropical plants, and most models achieve a good result in it. By the help of some tools and mobile apps allow farmers to upload leaf photos for real-time disease detection and it help them for their treatment recommendations. Draw back for this is real-world conditions, such as poor lighting and complex backgrounds, can lower accuracy. Identifying the exact stage of disease also remains a challenge. Overall, AI-based plant disease detection is a quick and reliable solution to reduce crop loss, increases yeild and promote smart agriculture.

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